

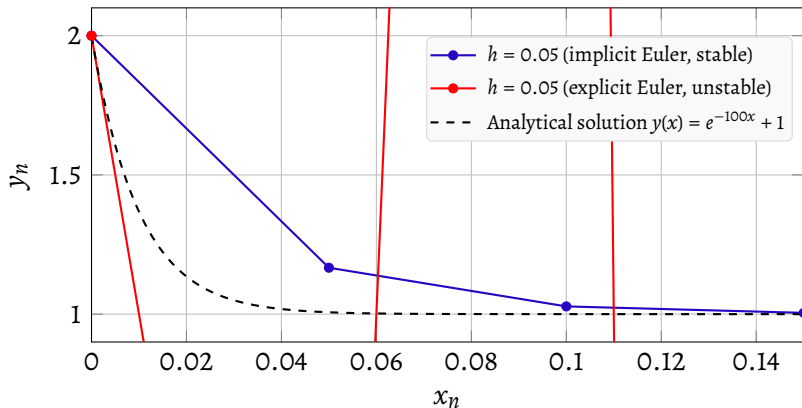


Figure 20 – Numerical stiffness in the equation $y' = -100y + 100$, $y(0) = 2$, with the same step $h = 0.05$ in two distinct methods. The explicit Euler method (red curve) is numerically unstable and generates values that blow up in magnitude and switch sign at each iteration. The implicit Euler method (blue curve), on the other hand, remains stable and approaches $y = 1$ rapidly, in agreement with the analytical solution (dashed curve). This contrast shows why stiff problems require implicitly stable methods.

